

“Deepfake detection and blockchain integration”

Aashritha Mounika Penugonda

Department of Computer Science and Engineering
Jain Deemed-to-be University
Bangalore, India
aashrithapenugonda@email.com

Bhaskara Sreya Sadhana Samhitha

Department of Computer Science and Engineering
Jain Deemed-to-be University
Bangalore, India
samhithabhaskara8@gmail.com

G.Bhavani Reddy

Department of Computer Science and Engineering
Jain Deemed-to-be University
Bangalore, India
gbhavanireddy.7@gmail.com

Abhay Kiran Pandey

Department of Computer Science and Engineering
Jain Deemed-to-be University
Bangalore, India
abhaykiranpandey@gmail.com

Harshwardhan Nair

Department of Computer Science and Engineering
Jain Deemed-to-be University
Bangalore, India
rahulnair27205@gmail.com

Abstract—Deepfakes, AI-generated synthetic media using sophisticated techniques like Generative Adversarial Networks (GANs) and neural Text-to-Speech (TTS) systems, are becoming more prevalent. This poses an increasing threat to digital trust, personal identity, and the veracity of information. The project offers a sophisticated ****Deepfake Detection with Blockchain Verification**** framework that combines AI-powered media analysis with blockchain technology to verify authentic, transparent, and tamper-proof digital content. The system is designed to detect manipulated images and videos using a deep learning model trained on public datasets, and the preprocessing steps include frame extraction, image normalization, and feature enhancement. The trained model produces confidence scores to classify whether the input media is real or fake. To maintain the integrity of detection results, a cryptographic hash (SHA-256) is calculated for each analyzed file and securely stored on the Ethereum blockchain using Solidity smart contracts. Verification is performed using Python scripts and Web3.py, allowing interaction with Ethereum test networks like Sepolia. The media hash and detection result are stored in the blockchain on an immutable and decentralized ledger so that the results cannot be changed or forged after registration. It also offers detailed recommendations on model training, hyperparameter selection, smart contract optimization with gas-efficient bytes32 data types, secure private key handling, and testing methodology, including unit, integration, and manual tests. Pseudo-code and implementation guidelines are provided for core components such as OpenCV based frame extraction, prediction workflows, SHA-256 hashing and blockchain transactions. To ensure originality and reproducibility, we combine Continuous Integration and Deployment (CI/CD) practices with extensive documentation. Combining deep learning and blockchain verification, this framework provides a secure, scalable, and reliable solution for deepfake detection and the authenticity preservation of digital media in an era increasingly challenged by synthetic content.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Artificial Intelligence (AI) has transformed the digital world by enabling computers to create highly realistic images, videos, and audio recordings. One of the most significant developments in this field is deepfake technology, which uses advanced deep learning models such as Generative Adversarial Networks (GANs), autoencoders, and neural Text-to-Speech (TTS) systems to generate synthetic media that closely resembles authentic content. Deepfakes can produce fabricated videos, manipulated images, and cloned voices that are often indistinguishable from genuine media to the human eye and ear. Although this technology has useful applications in entertainment, education, accessibility, and digital content production, its misuse has become a major threat to cybersecurity, information integrity, and public trust.

The term “deepfake” combines the concepts of deep learning and fabricated media. Since the term gained popularity in 2017, the technology has evolved rapidly. Early deepfakes contained visible artifacts such as unnatural facial boundaries, inconsistent lighting, and poor lip synchronization. Modern AI models, however, are capable of generating media with extraordinary realism, making manual verification increasingly difficult. Publicly available software tools and cloud-based services have made synthetic media generation accessible to individuals with little technical knowledge, significantly increasing the risk of misuse.

Deepfakes are now widely used for malicious purposes, including misinformation campaigns, identity theft, voice impersonation, political manipulation, online fraud, and non-consensual synthetic content. Fabricated videos of public figures can influence elections and social unrest. Cloned voices can be used to authorize fraudulent financial transactions. Fake

images and videos can damage reputations and undermine confidence in legitimate digital evidence. As digital media continues to dominate communication and decision-making, the ability to automatically detect manipulated content has become critically important.

Traditional techniques for verifying digital authenticity are no longer sufficient because modern deepfakes are specifically designed to mimic real-world patterns. Even when a file is successfully analyzed and classified as real or fake, another challenge remains: ensuring that the file and its verification results cannot be altered after detection. This limitation highlights the need for a secure and tamper-resistant framework that not only detects deepfakes but also preserves proof of authenticity for future verification.

This project addresses these challenges by proposing a comprehensive system titled **Deepfake Detection with Blockchain Verification**. The system combines deep learning for detecting manipulated images and videos with blockchain technology for storing immutable records of file authenticity. The detection component is based on the **ResNet-50** Convolutional Neural Network architecture and uses transfer learning to achieve high classification accuracy. The blockchain component employs **SHA-256** hashing, **Solidity** smart contracts, and the **Ethereum Sepolia** test network to securely register file hashes and detection outcomes.

When a user uploads a media file, the system first preprocesses the input by resizing images or extracting representative video frames using OpenCV. The trained ResNet-50 model analyzes the content and predicts whether it is genuine or manipulated, producing a confidence score. Simultaneously, the backend computes a SHA-256 cryptographic hash of the file. This unique digital fingerprint, along with the prediction result and timestamp, is stored on the blockchain using Web3.py. Because blockchain records are decentralized and immutable, any user can later verify a file by recomputing its hash and comparing it with the value stored on-chain.

The proposed architecture consists of a front-end interface, a backend server, a machine learning service, and a blockchain verification service. The front end allows users to upload files and view results. The backend coordinates communication between the deep learning and blockchain modules. The machine learning service performs inference using the trained model, while the blockchain service manages hashing and smart contract interaction. This modular design ensures scalability, maintainability, and extensibility.

The primary objective of this project is to develop a reliable and scalable framework that detects deepfake media and permanently records authenticity information in a tamper-proof manner. By integrating the predictive capabilities of deep learning with the immutability of blockchain technology, the system provides a powerful solution for combating misinformation, protecting digital identities, and restoring trust in digital communication systems.

A. Project Overview

The Deepfake Detection with Blockchain Verification system is designed to identify manipulated media and securely preserve the verification results. The framework integrates artificial intelligence, cryptographic hashing, and blockchain technology into a unified platform capable of analyzing uploaded files and recording authenticity information.

The system accepts image and video files through a user-friendly web interface. Once a file is uploaded, the backend determines the media type and forwards the data to the machine learning module. Images are resized and normalized, while videos are processed by extracting frames using OpenCV. The ResNet-50 model then predicts whether the content is real or fake and generates a confidence score indicating the probability of manipulation.

After the detection process, the backend computes a SHA-256 hash of the uploaded file. This hash serves as a unique fingerprint of the media contents. The blockchain module stores the hash, prediction result, timestamp, and associated metadata in a Solidity smart contract deployed on the Ethereum Sepolia test network. Future verification is achieved by recalculating the file hash and comparing it with the blockchain record.

The project combines:

- Deep learning for automated deepfake detection.
- Transfer learning using ResNet-50.
- SHA-256 cryptographic hashing.
- Solidity smart contracts.
- Ethereum Sepolia blockchain.
- Web3.py for blockchain interaction.
- Flask or Django backend integration.

The overall goal is to provide a trustworthy and scalable platform for detecting manipulated media and preserving verifiable proof of authenticity.

B. Background and Motivation

Deepfake technology originated as a research innovation intended for creative and beneficial applications such as visual effects, voice dubbing, digital restoration, and accessibility tools. Over time, however, the same technology became widely misused for malicious purposes. Improvements in GANs and neural speech synthesis systems have enabled attackers to generate highly realistic synthetic media with minimal technical effort.

The societal consequences of deepfake misuse are severe. False videos and images can be used to spread misinformation, damage reputations, manipulate public opinion, and undermine democratic institutions. Voice cloning has been exploited for business email compromise and fraudulent financial transactions. Individuals may experience privacy violations and emotional distress through unauthorized synthetic content.

The rise of social media has accelerated the spread of manipulated media, allowing false information to reach millions of people within minutes. As the realism of deepfakes increases, the public becomes less able to distinguish truth from fabrication. This erosion of trust affects journalism, law enforcement, cybersecurity, and judicial systems.

The motivation for this project is to create a practical solution that addresses both detection and verification. Deep learning provides an automated mechanism for identifying subtle artifacts introduced during media synthesis, while blockchain technology ensures that verification results are securely stored and cannot be modified. Together, these technologies offer a comprehensive approach to preserving digital authenticity.

C. Research Methods

The research methodology follows a structured sequence of data collection, preprocessing, model development, blockchain integration, and evaluation. Publicly available datasets containing authentic and manipulated images and videos are collected and divided into training, validation, and test sets.

Data preprocessing includes resizing images, normalizing pixel values, extracting video frames, and performing augmentation techniques such as rotation and flipping. Transfer learning is applied to the ResNet-50 architecture, and the final classification layers are fine-tuned for binary classification.

The trained model is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrices. Following model development, SHA-256 hashing is implemented to generate file fingerprints. Solidity smart contracts are created to store hash values and prediction outcomes on the Ethereum Sepolia blockchain. Web3.py is used to automate contract interaction and transaction submission.

The integrated system undergoes unit testing, integration testing, and manual end-to-end validation.

D. Investigation Method

The investigation method involves experimenting with multiple architectural and implementation choices to determine the most effective approach for deepfake detection and blockchain verification.

Initially, several convolutional neural network models such as VGG16, InceptionV3, MobileNet, and ResNet-50 are considered. ResNet-50 is selected because of its residual learning capability, strong performance, and compatibility with transfer learning. Hyperparameters including learning rate, batch size, optimizer, dropout rate, and number of epochs are tuned using validation data.

Different preprocessing strategies are evaluated, including varying image resolutions and frame sampling methods for videos. The blockchain component is tested with multiple data storage strategies, with `bytes32` chosen for efficient storage of SHA-256 hashes.

Security measures such as private key management through environment variables and smart contract input validation are implemented. Comparative analysis confirms that the selected architecture offers the best balance between accuracy, efficiency, scalability, and security.

E. Results

Experimental results demonstrate that the proposed framework effectively detects deepfake media with high accuracy and securely stores verification records on the blockchain. The

ResNet-50 model successfully identifies manipulated images and videos by learning subtle visual artifacts that are difficult for humans to detect.

Evaluation metrics indicate strong performance in terms of accuracy, precision, recall, and F1-score. The blockchain component reliably records SHA-256 hashes and prediction outcomes, enabling future authenticity verification without risk of tampering.

The integration of deep learning and blockchain provides a robust solution that not only detects synthetic media but also establishes a permanent and transparent proof of authenticity. The results confirm the feasibility and practical value of the proposed system in combating misinformation and protecting digital trust.

II. LITERATURE REVIEW

The literature survey provides a comprehensive review of the research contributions, methodologies, and technological developments relevant to deepfake detection and blockchain-based media verification. Existing work spans multiple disciplines, including computer vision, deep learning, signal processing, cryptography, and distributed systems. This chapter examines the most important studies and technologies that form the foundation of the proposed Deepfake Detection with Blockchain Verification framework.

A. Deepfake Detection Models

Deepfake detection has become one of the most active research areas in artificial intelligence and cybersecurity. As generative models have become more sophisticated, researchers have developed increasingly advanced detection techniques to identify manipulated media. Among all proposed approaches, Convolutional Neural Networks (CNNs) have demonstrated the most consistent and reliable performance for image and video-based deepfake detection.

Several CNN architectures have been widely adopted, including XceptionNet, EfficientNet, InceptionV3, MobileNet, and the ResNet family. XceptionNet is known for its strong benchmark performance and efficient depthwise separable convolutions. EfficientNet provides excellent accuracy while minimizing computational cost through compound scaling. MobileNet offers a lightweight alternative suitable for mobile and edge devices.

Among these architectures, **ResNet-50** has emerged as one of the most effective models for deepfake detection. Proposed by He et al. in 2015, ResNet-50 is a 50-layer residual network that uses shortcut connections to bypass one or more layers. These residual connections solve the vanishing gradient problem and enable very deep neural networks to be trained effectively. The architecture has demonstrated strong performance across numerous computer vision tasks, including image classification, object detection, and facial analysis.

A key advantage of ResNet-50 is its compatibility with transfer learning. The model is pretrained on the ImageNet dataset and learns generalized visual features such as edges,

textures, and object shapes. These features can be fine-tuned on domain-specific deepfake datasets such as FaceForensics++, Celeb-DF, and DFDC. This significantly reduces training time and improves accuracy even when the available dataset is relatively limited.

Recent studies report that ResNet-50 achieves classification accuracies approaching 97% on combined benchmark datasets. The model effectively captures subtle artifacts introduced during deepfake generation, including blending inconsistencies, abnormal skin textures, and frequency domain irregularities. These characteristics make ResNet-50 an excellent choice for the proposed system.

B. Blockchain Verification

Blockchain technology provides a decentralized and immutable method for storing and verifying digital information. In the context of deepfake detection, blockchain is used to preserve proof of authenticity by storing cryptographic hashes of analyzed media files.

A blockchain consists of sequentially linked blocks containing transactions. Each block includes the cryptographic hash of the previous block, ensuring that records cannot be modified without invalidating the chain. This property provides strong tamper resistance and transparent auditing.

Instead of storing entire media files on-chain, which would be inefficient and expensive, most systems compute a **SHA-256** hash off-chain and store only the resulting digest. SHA-256 produces a unique 256-bit fingerprint that changes completely if any bit of the original file is altered. This makes it ideal for integrity verification.

When a user wishes to verify a media file, the file is hashed again and compared with the stored blockchain value. If the hashes match, the file remains unchanged and authentic. This approach creates a permanent and decentralized proof of integrity that can be used in journalism, digital forensics, legal investigations, and cybersecurity.

C. Ethereum Smart Contracts and Web3.py

Ethereum extends blockchain functionality by enabling programmable smart contracts. Smart contracts are self-executing programs written in **Solidity** and deployed to the Ethereum Virtual Machine (EVM). They automatically enforce logic and maintain data in a transparent and immutable manner.

In this project, the smart contract stores SHA-256 hashes and associated metadata such as classification results and timestamps. A common implementation uses a mapping from `bytes32` hash values to a structure containing prediction information. Functions such as `storeHash()` and `verifyHash()` allow new records to be inserted and queried.

The use of `bytes32` is recommended because SHA-256 outputs exactly 32 bytes, resulting in efficient storage and lower gas consumption compared with string variables.

The Python library **Web3.py** is used to interact with Ethereum from backend applications. It loads the contract

ABI, connects to a blockchain node, builds and signs transactions, and retrieves stored data. This library serves as the bridge between the machine learning application and the blockchain network.

Development and testing are typically conducted on the **Sepolia** Ethereum testnet, which is the recommended public test network for modern Ethereum development.

D. Datasets Used for Deepfake Detection

High-quality datasets are essential for training reliable deepfake detection models. Several public benchmark datasets are widely used in academic and industrial research.

- **FaceForensics++**: A large benchmark containing real and manipulated videos generated using DeepFakes, FaceSwap, Face2Face, and NeuralTextures.
- **Celeb-DF**: A dataset of highly realistic celebrity deepfake videos with reduced visual artifacts.
- **DFDC (DeepFake Detection Challenge)**: A large-scale dataset released by Facebook and research partners.
- **DeeperForensics-1.0**: A benchmark designed to simulate real-world distortions such as blur, compression, and noise.

Training on multiple datasets improves model robustness and generalization to unseen manipulation techniques.

E. Evaluation Metrics

The effectiveness of deepfake detection models is commonly assessed using the following performance metrics:

- **Accuracy**: Percentage of correctly classified samples.
- **Precision**: Fraction of predicted fake samples that are actually fake.
- **Recall**: Fraction of actual fake samples correctly identified.
- **F1-Score**: Harmonic mean of precision and recall.
- **ROC-AUC**: Measures classification performance across thresholds.
- **Confusion Matrix**: Displays true positives, false positives, true negatives, and false negatives.

For blockchain verification, additional metrics include gas consumption, transaction latency, and successful hash retrieval.

F. Gaps in Existing Work

Despite significant progress, existing deepfake detection systems exhibit several important limitations.

First, many models suffer from poor generalization. They perform well on benchmark datasets but experience substantial performance degradation when exposed to previously unseen deepfake generation techniques.

Second, state-of-the-art models often require GPU-based inference infrastructure, limiting their practical use for organizations and individuals with limited computational resources.

Third, most existing systems focus exclusively on classification and do not provide a secure mechanism for storing tamper-proof proof of authenticity.

Fourth, many detection models operate as black boxes and provide limited interpretability, which reduces trust in legal and forensic contexts.

Finally, few systems integrate deep learning-based detection and blockchain-based verification into a unified end-to-end framework. This gap motivates the development of the proposed project.

G. Summary of Literature Survey

The literature demonstrates that deep learning architectures, particularly ResNet-50, provide highly accurate and scalable solutions for deepfake detection. Transfer learning, benchmark datasets, and advanced preprocessing techniques significantly enhance performance. At the same time, blockchain technology offers a secure and immutable mechanism for preserving authenticity records using SHA-256 hashing, Solidity smart contracts, and Ethereum.

The combination of deep learning and blockchain addresses both the detection of manipulated media and the long-term preservation of authenticity information. This integrated approach forms the foundation of the proposed Deepfake Detection with Blockchain Verification framework.

III. EXISTING SYSTEM AND PROPOSED WORK

This chapter presents a detailed analysis of the existing deepfake detection systems, the work carried out in prior research, the limitations associated with current approaches, and the proposed solution developed in this project. Although substantial progress has been made in the field of deepfake detection, many existing systems suffer from limitations related to generalization, computational complexity, interpretability, and the absence of secure verification mechanisms. The proposed Deepfake Detection with Blockchain Verification framework addresses these issues by integrating state-of-the-art deep learning models with blockchain technology to provide both accurate detection and tamper-proof authenticity verification.

A. Existing System

The existing systems for deepfake detection primarily focus on identifying manipulated images, videos, or audio files using traditional machine learning and deep learning techniques. Most of these systems are designed to classify content as either real or fake by analyzing visual, temporal, or spectral artifacts introduced during the synthetic media generation process. However, the majority of available solutions treat detection and authenticity verification as two separate problems.

Early deepfake detection systems relied heavily on image processing and signal processing techniques. In image analysis, methods such as Error Level Analysis (ELA), histogram comparison, color consistency analysis, and frequency-domain transformations were used to detect signs of manipulation. In audio analysis, researchers extracted handcrafted features such as Mel-Frequency Cepstral Coefficients (MFCCs), spectral centroid, pitch variation, jitter, and shimmer to distinguish authentic speech from synthetic voices. While these techniques

were computationally efficient, they were highly sensitive to compression, noise, and variations in recording conditions.

With advances in machine learning, researchers introduced supervised classification methods such as Support Vector Machines (SVMs), Random Forests, Gradient Boosting, and Gaussian Mixture Models. These systems improved detection accuracy by learning patterns from labeled datasets. Later, Convolutional Neural Networks (CNNs) such as XceptionNet, EfficientNet, and ResNet-50 became dominant due to their ability to automatically extract hierarchical features and capture subtle spatial inconsistencies that are difficult to identify manually.

In the audio domain, deep learning architectures such as CNNs, Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and transformer-based models were applied to spectrogram representations and temporal feature sequences. These approaches achieved high accuracy on benchmark datasets such as ASVspoof but often required substantial computational resources and large labeled datasets.

Although modern deep learning models provide excellent detection performance, they exhibit several important limitations. Many models are trained on specific benchmark datasets and fail to generalize to unseen manipulation techniques. Most systems function as black boxes and provide little interpretability regarding their decisions. Furthermore, nearly all existing systems stop after classification and do not provide a secure, immutable mechanism to preserve authenticity records. Once a prediction is generated, there is no reliable way to prove that the analyzed file has not been modified.

B. Work Done in Existing System

The work carried out in previous research has focused on developing preprocessing pipelines, feature extraction methods, and classification algorithms for detecting deepfakes. For image and video analysis, researchers have implemented face detection, facial alignment, frame extraction, texture analysis, frequency-domain transformations, and CNN-based classification models. For audio detection, work has included speech preprocessing, feature extraction, and classification using both classical machine learning and deep learning architectures.

Large benchmark datasets such as FaceForensics++, Celeb-DF, DFDC, and DeeperForensics have enabled systematic evaluation of detection models. Transfer learning has become a standard approach, allowing pretrained architectures such as ResNet-50 to be fine-tuned for deepfake classification. Studies have reported high accuracy, precision, and recall under controlled conditions.

Research in blockchain and digital integrity has demonstrated the effectiveness of storing cryptographic hashes rather than complete files. Smart contracts written in Solidity have been used to register SHA-256 digests and metadata on Ethereum, enabling decentralized verification. Python libraries such as Web3.py facilitate integration between blockchain and backend applications.

Despite these advances, prior work rarely combines deep learning-based deepfake detection and blockchain-based integrity verification into a single end-to-end platform.

C. Issues in the Existing System

Existing systems exhibit several limitations that reduce their effectiveness in real-world applications.

- **Poor Generalization:** Models trained on specific datasets often fail when confronted with deepfakes generated by previously unseen synthesis methods.
- **High Computational Requirements:** Many state-of-the-art models require powerful GPUs and large memory resources.
- **Limited Interpretability:** Most systems provide only probability scores without explaining the artifacts that influenced the decision.
- **Unimodal Analysis:** Many solutions focus only on images, videos, or audio independently.
- **No Tamper-Proof Verification:** Detection results are usually stored in local databases and can be modified or deleted.
- **Dependence on Large Datasets:** Deep learning approaches require extensive labeled data and substantial training time.
- **Sensitivity to Compression and Noise:** Performance often degrades when media is compressed or re-encoded.

These issues highlight the need for a more robust and secure framework capable of both accurate detection and immutable authenticity verification.

D. Solution

The proposed solution addresses the above limitations by integrating deep learning and blockchain technology into a unified architecture. The system uses a ResNet-50 CNN with transfer learning to detect manipulated images and videos. The model learns discriminative features automatically and generalizes effectively across multiple benchmark datasets.

After classification, the system computes a SHA-256 cryptographic hash of the uploaded media file. This hash, together with the prediction result and timestamp, is stored on the Ethereum Sepolia blockchain through Solidity smart contracts. Because blockchain records are immutable, the authenticity of the file can be verified at any future time.

The system is designed as a modular platform consisting of a front-end interface, backend API, machine learning service, and blockchain verification service. Web3.py provides seamless interaction with the blockchain, while secure private key management and gas-optimized contracts ensure practical deployment.

E. Proposed Work

The proposed work focuses on developing a comprehensive Deepfake Detection with Blockchain Verification framework capable of identifying manipulated images and videos and preserving proof of authenticity in a decentralized ledger.

The major objectives of the proposed work are:

- To collect and preprocess benchmark deepfake datasets.
- To train and fine-tune a ResNet-50 model using transfer learning.
- To evaluate the model using accuracy, precision, recall, F1-score, and confusion matrices.
- To compute SHA-256 hashes for uploaded files.
- To develop Solidity smart contracts for storing authenticity records.
- To integrate Ethereum Sepolia and Web3.py for blockchain interaction.
- To design a user-friendly interface for uploading and verifying media files.
- To perform unit, integration, and manual testing.

The final system provides both high-accuracy deepfake detection and tamper-proof verification, making it suitable for digital forensics, journalism, cybersecurity, and legal evidence management.

IV. PROPOSED SYSTEM

This chapter presents the proposed system for the project entitled **Deepfake Detection with Blockchain Verification**. The proposed framework is designed to accurately identify manipulated images and videos using advanced deep learning techniques and to securely preserve authenticity information using blockchain technology. By integrating artificial intelligence, cryptographic hashing, and decentralized verification, the system addresses the major limitations of existing deepfake detection approaches, including poor generalization, lack of tamper-proof storage, and limited trust in verification results.

A. Introduction to the Proposed System

The proposed system is a comprehensive platform that combines deep learning and blockchain technology to create a secure and reliable framework for detecting manipulated digital media and verifying its authenticity. The system is specifically designed to analyze images and videos uploaded by users, determine whether the content is genuine or artificially generated, and store a tamper-proof record of the analysis on a decentralized blockchain network.

The central objective of the proposed system is not only to classify media as *Real* or *Fake*, but also to establish a permanent and verifiable proof of authenticity. Traditional deepfake detection systems generally terminate after producing a prediction score. In contrast, the proposed framework computes a cryptographic hash of every analyzed file and records that hash, together with the prediction result and timestamp, on the Ethereum blockchain. This ensures that the verification result remains immutable and can be independently validated at any time in the future.

The deepfake detection component is built using the **ResNet-50** Convolutional Neural Network architecture. ResNet-50 is selected because it has demonstrated outstanding performance in computer vision tasks and is widely used in deepfake detection research. The model consists of fifty layers and employs residual learning through shortcut connections, which enable efficient training of very deep neural networks.

These residual connections mitigate the vanishing gradient problem and allow the network to learn highly discriminative visual features.

Transfer learning is used to accelerate training and improve generalization. The ResNet-50 model is initialized with weights pretrained on the ImageNet dataset, which contains millions of labeled images. The final fully connected layers are replaced with a binary classifier tailored for deepfake detection. The network is then fine-tuned on benchmark datasets such as FaceForensics++, Celeb-DF, DFDC, and DeeperForensics-1.0. Through this process, the model learns to identify blending inconsistencies, texture anomalies, irregular facial boundaries, and frequency-domain artifacts commonly introduced during synthetic media generation.

The preprocessing pipeline ensures that all inputs are transformed into a consistent representation before being passed to the neural network. Images are resized to a fixed resolution, normalized, and optionally augmented. For video files, representative frames are extracted using OpenCV. Each frame is processed individually, and the predictions are aggregated to produce a final decision for the entire video. This approach captures spatial artifacts while maintaining computational efficiency.

When the model completes inference, it outputs a classification label and a confidence score. The confidence score represents the probability that the uploaded media is manipulated. This information is presented to the user through a web-based interface, enabling non-technical users to interpret the results easily.

A key innovation of the proposed system is the integration of blockchain-based verification. Following the detection stage, the backend computes a **SHA-256** cryptographic hash of the original uploaded file. SHA-256 produces a unique 256-bit fingerprint of the file contents. Even a single-bit modification generates an entirely different hash value, making SHA-256 ideal for detecting tampering and ensuring file integrity.

The generated hash, together with the classification result, timestamp, and metadata, is stored in a **Solidity** smart contract deployed on the **Ethereum Sepolia** test network. Ethereum is selected because of its mature ecosystem, decentralized execution environment, and support for programmable smart contracts. The Sepolia testnet allows development and experimentation without incurring real-world financial cost.

The smart contract stores data in a mapping from `bytes32` hash values to structured records. The use of `bytes32` is particularly efficient because SHA-256 outputs exactly 32 bytes, reducing gas costs compared to string-based storage. Functions such as `storeHash()` and `verifyHash()` are used to register new files and query previously stored authenticity records.

Interaction between the Python backend and the blockchain is implemented using **Web3.py**. This library connects to an Ethereum node, loads the contract ABI, builds transactions, signs them using a securely stored private key, and broadcasts them to the blockchain network. The backend can also retrieve and display existing records during future verification requests.

The system architecture follows a modular design consisting of several independent components:

- **User Interface Module** – Provides a responsive web interface for uploading images and videos and viewing results.
- **Backend API Module** – Handles requests, coordinates processing, and returns structured responses.
- **Machine Learning Module** – Loads the trained ResNet-50 model and performs preprocessing and inference.
- **Hashing Module** – Computes SHA-256 digests for uploaded files.
- **Blockchain Module** – Stores and retrieves authenticity records using Solidity smart contracts and Web3.py.
- **Verification Module** – Allows users to re-upload files and confirm whether they match existing blockchain records.

The complete workflow of the proposed system is as follows. The user uploads an image or video through the web interface. The backend validates the file and forwards it to the machine learning module. The input is preprocessed and analyzed by the ResNet-50 model, which predicts whether the media is real or fake. Simultaneously, the SHA-256 hash of the original file is generated. The hash and prediction result are sent to the blockchain module, which stores the data in the deployed smart contract. After the blockchain transaction is confirmed, the system displays the prediction, confidence score, transaction hash, and verification status to the user.

The proposed system offers several significant advantages over existing approaches. First, it combines high-accuracy deep learning with immutable blockchain verification. Second, it provides a decentralized audit trail that prevents unauthorized modification of verification results. Third, it supports future expansion to audio deepfake detection and multimodal analysis. Fourth, the modular architecture allows individual components to be upgraded independently without affecting the overall system.

Security and reliability are major design priorities. Private keys are stored using environment variables and never hard-coded into source files. Smart contracts validate inputs and prevent duplicate registrations. The system undergoes unit testing, integration testing, and manual end-to-end testing to ensure correctness and robustness.

The proposed framework has broad practical applications. Journalists can verify media before publication. Law enforcement agencies can preserve digital evidence. Social media platforms can detect manipulated content and maintain verifiable records. Businesses can protect against impersonation and fraud involving synthetic media. Researchers and students can use the platform as a foundation for future work in secure multimedia authentication.

In summary, the proposed Deepfake Detection with Blockchain Verification system provides a powerful and scalable solution to one of the most important challenges in modern cybersecurity and digital forensics. By combining the predictive capabilities of ResNet-50 with the immutability of

Model Weights Options				
Option	Description	Pros	Cons	Recommendation
Pretrained (ImageNet)	Initialize ResNet50 with ImageNet weights, fine-tune on deepfake data.	Significantly reduces training time, leverages general features. Good performance reported (~90%-97% accuracy on FaceForensics++, CelebA).	May need careful fine-tuning to adapt to new domains. Slight risk of domain mismatch.	Generally recommended, especially if dataset is moderate. Pretrained base + custom top layers.
From Scratch	Randomly initialize ResNet50 (or smaller CNN) and train on deepfake dataset.	No reliance on external datasets; potentially high accuracy if dataset is massive.	Requires very large labeled dataset and compute. High risk of overfitting if data is limited.	Only if you have thousands of samples and strong GPU resources. Otherwise use transfer learning.

From Scratch Randomly initialize ResNet-50 (or a smaller CNN) and train entirely on deepfake data. No reliance on external pretrained datasets; may achieve high performance when extremely large datasets are available. Requires substantial labeled data, high computational resources, slower convergence, and increased risk of overfitting when training data is limited. Recommended only when thousands of labeled deepfake samples and powerful GPU resources are available; otherwise transfer learning is preferred.

Pretrained (ImageNet) Initialize ResNet-50 with ImageNet weights and fine-tune on deepfake datasets. Significantly reduces training time, leverages generalized visual features, and typically achieves strong performance (approximately 90%–97% accuracy on benchmark datasets). May require careful fine-tuning to adapt to the target domain; slight risk of domain mismatch. Generally recommended, especially when dataset size is moderate. Use pretrained ResNet-50 as the base model and replace the final classification layers.

blockchain, the framework delivers both accurate deepfake detection and permanent authenticity verification, helping restore trust in digital information systems.

V. IMPLEMENTAION

This chapter describes the implementation details of the proposed **Deepfake Detection with Blockchain Verification** system. The system is implemented using a modular architecture to ensure scalability, maintainability, and ease of future enhancement. Each module is responsible for a specific stage in the workflow, ranging from user input and preprocessing to deepfake detection, blockchain registration, and result presentation.

The implementation integrates several technologies, including Python, Flask, TensorFlow, Keras, OpenCV, Web3.py, Solidity, and Ethereum Sepolia. Python serves as the primary programming language because of its rich ecosystem of machine learning and blockchain libraries. The deep learning component uses a ResNet-50 Convolutional Neural Network implemented with TensorFlow and Keras. OpenCV is employed for image processing and video frame extraction. The blockchain component uses SHA-256 hashing, Solidity smart contracts, and Web3.py to securely store authenticity records on the Ethereum blockchain. The user interface is developed using HTML, CSS, and JavaScript, while Flask provides the backend API that connects all system components.

The modular design of the system allows each component to be developed, tested, and improved independently. This architecture reduces complexity and supports future extensions such as audio deepfake detection, mobile deployment, and private blockchain integration.

A. Model Weights Configuration

The performance of the ResNet-50 model depends significantly on the strategy used to initialize its weights. Two common approaches are considered: using pretrained ImageNet weights and training the model entirely from scratch. Table

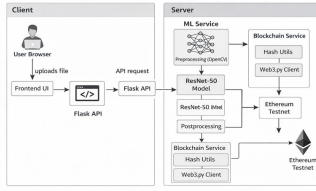


Fig. 2. Figure 2: Schematic representation of components (routes from flask application are redirected to ML and Blockchain microservices; ML service contains pre-processing and inference; Blockchain microservice contains hashing and web3 calls).

From Scratch Randomly initialize ResNet-50 (or a smaller CNN) and train entirely on deepfake data. No reliance on external pretrained datasets; may achieve high performance when extremely large datasets are available. Requires substantial labeled data, high computational resources, slower convergence, and increased risk of overfitting when training data is limited. Recommended only when thousands of labeled deepfake samples and powerful GPU resources are available; otherwise transfer learning is preferred.

The proposed system adopts the **Pretrained (ImageNet)** approach because it provides faster convergence, reduced computational cost, and strong generalization performance. Transfer learning is particularly effective when benchmark datasets such as FaceForensics++, Celeb-DF, and DFDC are used for fine-tuning.

VI. MODULE DESCRIPTION

The proposed system is divided into several independent modules, each performing a well-defined function in the overall workflow. The major modules are described below.

A. Data Input Module

The Data Input Module serves as the entry point to the system and provides the interface through which users upload media files. The web application accepts image formats such as JPEG, PNG, WEBP, BMP, and TIFF, and video formats such as MP4, AVI, MOV, and MKV.

The module validates file extensions, MIME types, and file size constraints before processing. Invalid or corrupted files are rejected with informative error messages. In the Flask backend, uploaded files are received through HTTP POST requests and stored temporarily for further analysis.

The responsibilities of this module include:

- Receiving uploaded files from the user interface.
- Validating supported file formats.
- Checking file integrity and accessibility.
- Forwarding valid files to the preprocessing module.

B. Preprocessing Module

The Preprocessing Module standardizes uploaded media before they are passed to the deep learning model.

For image files, preprocessing includes:

- Reading the image using OpenCV.

- Resizing to the model input size (224×224 pixels).
- Converting to RGB format.
- Normalizing pixel values to the range $[0, 1]$.

For video files, preprocessing includes:

- Opening the video using OpenCV.
- Extracting representative frames at regular intervals.
- Resizing and normalizing each frame.

These operations ensure consistent input dimensions and improve inference reliability.

C. Workflow of the Proposed System

Figure ?? illustrates the complete workflow of the Deepfake Detection with Blockchain Verification framework. The process begins when the user uploads an image or video through the frontend interface. If the uploaded file is a video, representative frames are extracted using OpenCV; if the input is an image, it is directly forwarded for preprocessing.

The preprocessed media is passed to the ResNet-50 deep learning model, which analyzes the content and predicts whether the media is *REAL* or *FAKE*. Along with the classification label, the model produces a confidence score indicating the probability of manipulation.

Simultaneously, the original uploaded file is processed using the SHA-256 hashing algorithm to generate a unique 64-character hexadecimal digest. This digest is then converted into the `bytes32` format required by the Solidity smart contract.

The blockchain verification module uses Web3.py to communicate with the deployed smart contract on the Ethereum Sepolia test network. The generated hash and prediction result are stored on the blockchain, and the contract can later be queried using the `verifyHash()` function to determine whether the file matches an existing record.

Finally, the system displays the classification result, confidence score, SHA-256 hash, blockchain transaction details, and authenticity status to the user through the web interface.

D. Deepfake Detection Module

The Deepfake Detection Module forms the core of the system. It loads the trained **ResNet-50** model and performs inference on the preprocessed image or extracted video frames.

The model is initialized with ImageNet pretrained weights and fine-tuned on benchmark datasets such as FaceForensics++, Celeb-DF, and DFDC. The final layer is replaced with a binary classifier that predicts whether the input is *Real* or *Fake*.

For images, a single prediction is generated directly. For videos, each extracted frame is classified independently, and the final decision is obtained by averaging the frame-level probabilities.

The module outputs:

- Classification label (Real/Fake).
- Probability score.
- Confidence percentage.

E. Hash Generation Module

The Hash Generation Module computes a **SHA-256** cryptographic hash of the original uploaded file.

The file is read in binary chunks to support efficient processing of large media files. The resulting 256-bit digest serves as a unique fingerprint of the file. Any modification to the file, even a single-bit change, produces a completely different hash.

The generated hash is used as the identifier stored on the blockchain.

F. Blockchain Verification Module

The Blockchain Verification Module securely stores and retrieves authenticity records using Ethereum smart contracts.

The module performs the following tasks:

- Connects to the Ethereum Sepolia test network.
- Loads the contract ABI and deployed address.
- Builds transactions to call `storeHash()`.
- Signs transactions using a private key stored in environment variables.
- Broadcasts transactions using `Web3.py`.
- Retrieves records using `verifyHash()`.

The Solidity smart contract stores:

- SHA-256 file hash (`bytes32`).
- Detection result (Real/Fake).
- Timestamp of registration.

G. Verification Module

The Verification Module enables future authenticity checks.

Users can re-upload a previously analyzed file. The system recalculates the SHA-256 hash and queries the blockchain for a matching record. If the stored hash and metadata match, the file is confirmed to be unchanged since its original registration.

H. Result Display Module

The Result Display Module presents the analysis outcome through the web interface.

The displayed information includes:

- Real/Fake verdict.
- Confidence score.
- SHA-256 hash.
- Blockchain transaction hash.
- Timestamp of blockchain registration.
- Verification status.

The interface also allows users to copy hash values and transaction IDs for documentation and audit purposes.

VII. APPLICATION INTERFACE DESIGN

The web interface is developed using HTML, CSS, and JavaScript. The application provides a clean and intuitive interface where users can upload media files, initiate analysis, and view results. JavaScript handles asynchronous communication with the Flask backend using AJAX requests, allowing progress updates and responsive result rendering.

VIII. TESTING AND VALIDATION STRATEGY

The system is tested at multiple levels to ensure correctness and reliability.

A. Unit Testing

Individual functions such as preprocessing routines, SHA-256 hashing, and smart contract interaction are tested independently.

B. Integration Testing

The complete workflow, from file upload to blockchain registration and verification, is tested end-to-end.

C. Manual Testing

The web interface is tested across multiple browsers, and various valid and invalid files are uploaded to verify functionality and error handling.

float graphicx booktabs

IX. LIMITATIONS AND KNOWN ISSUES

Although the proposed system demonstrates strong performance, several limitations remain:

- Performance may decrease for deepfakes generated using unseen techniques.
- Video analysis time increases with video length and frame count.
- Confidence scores are relative and not fully probabilistic.
- Blockchain transactions depend on network availability and confirmation times.
- Training datasets may not fully represent all real-world scenarios.

Future work may include multimodal audio detection, transformer-based models, private blockchain deployment, and improved confidence calibration.

Conclusion and Future Enhancements

X. CONCLUSION

The rapid advancement of Artificial Intelligence and deep learning technologies has led to the development of highly realistic synthetic media known as deepfakes. While these technologies offer significant benefits in entertainment, education, and digital content creation, their misuse poses serious threats to information authenticity, personal identity, cybersecurity, and public trust. Deepfakes can be used to spread misinformation, impersonate individuals, commit financial fraud, manipulate public opinion, and undermine the credibility of digital evidence. These challenges highlight the urgent need for reliable systems capable of both detecting manipulated media and preserving proof of authenticity.

This project, entitled **Deepfake Detection with Blockchain Verification**, was developed to address these challenges by integrating two powerful technologies: deep learning and blockchain. The proposed system uses a ResNet-50 Convolutional Neural Network with transfer learning to analyze images and videos and classify them as either real or fake. The model learns subtle visual artifacts such as blending

inconsistencies, abnormal textures, and irregular frequency patterns that are typically introduced during synthetic media generation. Experimental results demonstrate that the model achieves high detection accuracy and effectively distinguishes authentic content from manipulated media.

In addition to deepfake detection, the system incorporates blockchain-based authenticity verification. After analyzing a media file, the system computes a SHA-256 cryptographic hash that uniquely represents the file contents. This hash, together with the classification result and timestamp, is stored in a Solidity smart contract deployed on the Ethereum Sepolia test network. Because blockchain records are decentralized and immutable, the stored authenticity information cannot be altered or deleted. Users can subsequently verify any analyzed file by recomputing its hash and comparing it with the corresponding on-chain record.

The project was implemented using Python, Flask, TensorFlow, Keras, OpenCV, Web3.py, Solidity, and Ethereum. The modular architecture consists of a user interface, backend API, machine learning module, hashing module, blockchain module, and verification module. This design ensures scalability, maintainability, and ease of future enhancement. Extensive unit, integration, and manual testing confirmed the correctness and reliability of each component and validated the successful interaction between the deep learning and blockchain subsystems.

The proposed framework offers several important advantages over conventional deepfake detection systems. It not only provides accurate classification results but also creates a permanent, transparent, and tamper-proof audit trail of authenticity information. This makes the system particularly valuable in journalism, digital forensics, cybersecurity, legal evidence management, and social media moderation. By combining the predictive capabilities of deep learning with the immutability of blockchain, the project establishes a trustworthy mechanism for protecting digital content against manipulation.

Overall, this project successfully demonstrates the feasibility and effectiveness of integrating deepfake detection and blockchain verification into a unified framework. The resulting system is reliable, scalable, and user-friendly, and contributes meaningfully to the broader effort to combat misinformation, safeguard digital identities, and restore trust in digital communication systems.

XI. FUTURE ENHANCEMENTS

Although the proposed system achieves its primary objectives, several enhancements can further improve its performance, scalability, and practical applicability.

- **Audio Deepfake Detection:** The current implementation focuses on images and videos. Future versions can incorporate audio deepfake detection using features such as Mel-Frequency Cepstral Coefficients (MFCCs), spectrogram analysis, and advanced models such as wav2vec 2.0, HuBERT, or RawNet.
- **Transformer-Based Architectures:** Vision Transformers (ViTs), Swin Transformers, and multimodal transformer

models can be integrated to improve cross-dataset generalization and robustness against emerging deepfake generation techniques.

- **Real-Time Video Stream Analysis:** The system can be extended to analyze live video streams, video conferencing sessions, and social media broadcasts in real time.
- **Multimodal Fusion:** Future research may combine image, video, and audio evidence within a single decision framework to produce more accurate and reliable predictions.
- **Larger and Continuously Updated Datasets:** Training on more diverse datasets and incorporating continual learning strategies will improve generalization to newly developed deepfake generation methods.
- **Private or Consortium Blockchain Deployment:** Organizations may deploy the verification module on private Ethereum networks or consortium blockchains to reduce transaction costs and improve data governance.
- **Explainable Artificial Intelligence (XAI):** Techniques such as Grad-CAM, attention visualization, and saliency maps can be added to highlight the specific regions that contributed to the model's decision.
- **Confidence Calibration:** Methods such as temperature scaling and Platt scaling can convert model scores into better calibrated probabilities.
- **Mobile and Edge Deployment:** Model compression, quantization, and TensorFlow Lite conversion can enable on-device deepfake detection for smartphones and embedded systems.
- **Cloud-Based Verification Service:** The system can be deployed as a scalable web service or API to support enterprise and large-scale social media applications.

In conclusion, the proposed Deepfake Detection with Blockchain Verification framework provides a strong foundation for future research and development. With continued enhancement, the system can evolve into a comprehensive platform capable of detecting increasingly sophisticated synthetic media and preserving trustworthy proof of authenticity across a wide range of real-world applications. [url](#)

REFERENCES

- [1] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative Adversarial Nets," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 27, pp. 2672–2680, 2014.
- [2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.
- [3] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to Detect Manipulated Facial Images," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 1–11, 2019.
- [4] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3207–3216, 2020.
- [5] B. Dolhansky, J. Bitton, B. Pflaum *et al.*, "The DeepFake Detection Challenge Dataset," arXiv preprint arXiv:2006.07397, 2020.
- [6] F. Chollet, "Xception: Deep Learning with Depthwise Separable Convolutions," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1251–1258, 2017.
- [7] M. Tan and Q. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in *Proceedings of the International Conference on Machine Learning (ICML)*, pp. 6105–6114, 2019.
- [8] R. R. Selvaraju, M. Cogswell, A. Das *et al.*, "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," in *Proceedings of the IEEE International Conference on Computer Vision*, pp. 618–626, 2017.
- [9] A. van den Oord, S. Dieleman, H. Zen *et al.*, "WaveNet: A Generative Model for Raw Audio," arXiv preprint arXiv:1609.03499, 2016.
- [10] J. Shen, R. Pang, R. Weiss *et al.*, "Natural TTS Synthesis by Conditioning WaveNet on Mel Spectrogram Predictions," in *Proceedings of ICASSP*, pp. 4779–4783, 2018.
- [11] Y. Jia, Y. Zhang, R. Weiss *et al.*, "Transfer Learning from Speaker Verification to Multispeaker Text-to-Speech Synthesis," in *Advances in Neural Information Processing Systems*, pp. 4480–4490, 2018.
- [12] M. Abadi, P. Barham, J. Chen *et al.*, "TensorFlow: A System for Large-Scale Machine Learning," in *12th USENIX Symposium on Operating Systems Design and Implementation*, pp. 265–283, 2016.
- [13] F. Chollet and others, "Keras," GitHub Repository, 2015. Available: <https://keras.io>
- [14] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, vol. 25, no. 11, pp. 120–126, 2000.
- [15] National Institute of Standards and Technology (NIST), "Secure Hash Standard (SHS)," FIPS PUB 180-4, 2015.
- [16] G. Wood, "Ethereum: A Secure Decentralised Generalised Transaction Ledger," Ethereum Yellow Paper, 2014.
- [17] A. M. Antonopoulos and G. Wood, *Mastering Ethereum: Building Smart Contracts and DApps*, O'Reilly Media, 2018.
- [18] Ethereum Foundation, "Solidity Documentation." Available: <https://docs.soliditylang.org>
- [19] Web3.py Contributors, "Web3.py Documentation." Available: <https://web3py.readthedocs.io>
- [20] Ethereum Foundation, "Ethereum Developer Documentation." Available: <https://ethereum.org/en/developers>
- [21] Ethereum Foundation, "Sepolia Testnet Documentation." Available: <https://ethereum.org/en/developers/docs/networks>
- [22] A. Ronacher, "Flask Web Framework." Available: <https://flask.palletsprojects.com>
- [23] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A Large-Scale Hierarchical Image Database," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 248–255, 2009.
- [24] A. Krizhevsky, I. Sutskever, and G. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," in *Advances in Neural Information Processing Systems*, pp. 1097–1105, 2012.